

The Sharp Universal Constant in the Modified Log-Sobolev–Beckner Comparison

Candidate full solution for arXiv:2007.10209

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Status. Candidate full proof, likely valid, pending human review. Under Assumption 1 of the source, the optimal universal constant is exactly $K_p^{\text{opt}} = 1/2$ for every $p \in (1, 2]$.

1 Source and answer

R. Adamczak, B. Polaczyk, and M. Strzelecki, *Modified log-Sobolev inequalities, Beckner inequalities and moment estimates*, arXiv:2007.10209 [1], ask for the best universal constant K_p^{opt} in the implication

$$\rho_0 \text{Ent}_\mu(f) \leq \mathcal{E}(f, \log f) \implies \alpha_p(\mu(f^p) - \mu(f)^p) \leq \frac{p}{2} \mathcal{E}(f, f^{p-1}), \quad \alpha_p \geq K_p^{\text{opt}} \rho_0.$$

Their setting is a probability space with a symmetric nonnegative bilinear form satisfying their Assumption 1: contractions preserve its domain, and pointwise comparison of products of increments implies comparison of the corresponding energies.

$$k(p, \theta) = \frac{\mathcal{E}(f, f^p) - \mathcal{E}(f, f)^p}{\mathcal{E}(f, f^{p-1})}, \quad \theta \in (0, 1)$$

It is easy to see that $\lim_{p \rightarrow 2^-} K_p = 1/2$. Since $\lim_{p \rightarrow 1^+} k(p, (p-1)^2) = 1$ and obviously $k(p, \theta) \leq 1$, we conclude that $\lim_{p \rightarrow 1^+} K_p = 1/2$.

Moreover one can check that

$$(2.4) \quad K_p \geq \max\{1 - 1/p; p/2 \cdot k(p, 0.25 \cdot (1-p)^2)\} \geq 0.17.$$

This ends the proof of the theorem. $\square$

Remark 2.10. Note that the numerical bound (2.4) cannot be substantially improved if we want it to hold for *all* $p \in (1, 2]$. Indeed, we have

$$K_{6/5} \leq 0.18.$$

Identification of the best constants K_p^{opt} such that $\alpha_p \geq K_p^{\text{opt}} \rho_0$ seems to be an interesting open question.

3. MOMENT ESTIMATES DERIVED FROM BECKNER'S INEQUALITIES

In this section we revisit the arguments by Boucheron et al. [27, Theorem 15.5] and present them in the context of general Beckner inequalities. We derive moment inequalities, which are valid in particular under the assumption of mod-

Figure 1: The exact open question in the source, PDF page 12.

The answer is independent of p .

2 Proof intuition

Power entropy is a positive Stieltjes mixture of logarithmic entropies of the shifts $f + s$. Applying mLSI to every shift and integrating converts the entropy side exactly into $\mu(f^p) - \mu(f)^p$ and the energy side exactly into $\frac{p}{2} \mathcal{E}(f, f^{p-1})$, giving the universal factor 1/2. The symmetric two-point form attains that ratio, so no larger universal constant is possible.

Theorem 1. *In the abstract setting of the source, for every $p \in (1, 2]$,*

$$\alpha_p^{\text{opt}} \geq \frac{\rho_0^{\text{opt}}}{2}.$$

The factor $1/2$ is attained by the symmetric two-point space. Consequently,

$$\boxed{K_p^{\text{opt}} = \frac{1}{2} \quad (1 < p \leq 2).}$$

3 A positive mixture of shifted logarithmic entropies

Fix $1 < p < 2$, put $r = p - 1 \in (0, 1)$, and define

$$C_p = \frac{\pi}{p(p-1) \sin(\pi(p-1))}.$$

The elementary Stieltjes identity

$$\int_0^\infty \frac{s^{p-2}}{x+s} ds = \frac{\pi x^{p-2}}{\sin(\pi(p-1))}, \quad x > 0, \quad (1)$$

is the main point. For $\phi_s(x) = (x+s) \log(x+s)$, integrate the Bregman remainder of ϕ_s and use (1) twice. For all $x, m \geq 0$,

$$\begin{aligned} \int_0^\infty s^{p-2} [\phi_s(x) - \phi_s(m) - \phi'_s(m)(x-m)] ds \\ = C_p [x^p - m^p - pm^{p-1}(x-m)]. \end{aligned} \quad (2)$$

Thus, for every bounded nonnegative f ,

$$\int_0^\infty s^{p-2} \text{Ent}_\mu(f+s) ds = C_p [\mu(f^p) - \mu(f)^p]. \quad (3)$$

All integrands in (2) are nonnegative, so Tonelli's theorem applies without subtracting divergent affine terms.

The companion identity, obtained by integrating (1) from b to a , is

$$\int_0^\infty s^{p-2} [\log(a+s) - \log(b+s)] ds = pC_p (a^{p-1} - b^{p-1}) \quad (4)$$

for $a, b \geq 0$, with the boundary cases understood by monotone limits.

4 Integrating mLSI

Let f be bounded and nonnegative, with $f, f^{p-1} \in \text{Dom}(\mathcal{E})$. For each $s > 0$, the shifted function $f+s$ belongs to the domain. Also $\log(f+s)$ belongs to it: on the range of f , this logarithm is Lipschitz, and a scaled Lipschitz extension is a contraction. Hence mLSI gives

$$\rho_0 \text{Ent}_\mu(f+s) \leq \mathcal{E}(f, \log(f+s)). \quad (5)$$

We record the minor domain point behind integrating (5). For $0 < \varepsilon < R < \infty$, set

$$H_{\varepsilon, R}(x) = \int_\varepsilon^R s^{p-2} \log(x+s) ds.$$

This function is Lipschitz on the range of f , so $H_{\varepsilon,R}(f) \in \text{Dom}(\mathcal{E})$. Positive Riemann sums of (5) converge to

$$\rho_0 \int_{\varepsilon}^R s^{p-2} \text{Ent}_{\mu}(f+s) ds \leq \mathcal{E}(f, H_{\varepsilon,R}(f)).$$

This passage does not require an unstated continuity of $\mathcal{E}$: the derivatives of the Riemann-sum kernels converge uniformly on the bounded range of f , and Assumption 1 squeezes their energies between $(1 \pm o(1))$ times the energy of $H_{\varepsilon,R}$.

For every $a, b \geq 0$, positivity and (4) give

$$\begin{aligned} (a-b)(H_{\varepsilon,R}(a) - H_{\varepsilon,R}(b)) \\ \leq pC_p(a-b)(a^{p-1} - b^{p-1}). \end{aligned}$$

The pointwise-comparison clause of Assumption 1 therefore yields

$$\mathcal{E}(f, H_{\varepsilon,R}(f)) \leq pC_p \mathcal{E}(f, f^{p-1}).$$

Letting $\varepsilon \downarrow 0$ and $R \uparrow \infty$, monotone convergence and (3) imply

$$\rho_0 [\mu(f^p) - \mu(f)^p] \leq p \mathcal{E}(f, f^{p-1}). \quad (6)$$

Comparing (6) with the source's normalization of the Beckner inequality gives $\alpha_p \geq \rho_0/2$.

The bounded restriction is harmless: the source's truncation lemma extends a Beckner inequality proved for bounded functions, with the same constant, to all functions in its stated domain. At $p = 2$, the conclusion follows by linearizing mLSI at $f = 1$: $\rho_0 \text{Var}_{\mu}(g)/2 \leq \mathcal{E}(g, g)$. This proves the lower bound in Theorem 1.

5 Sharpness on two points

Let $\mathcal{X} = \{0, 1\}$, let μ be uniform, and for $\gamma > 0$ define

$$\mathcal{E}_{\gamma}(u, v) = \gamma [u(1) - u(0)][v(1) - v(0)].$$

This form satisfies Assumption 1. We compute both optimal constants.

By homogeneity, write the two values of a positive function as $m(1+x), m(1-x)$, where $|x| < 1$. Then

$$\begin{aligned} \text{Ent}_{\mu}(f) &= m \sum_{n \geq 1} \frac{x^{2n}}{2n(2n-1)}, \\ \mathcal{E}_{\gamma}(f, \log f) &= 4\gamma m \sum_{n \geq 1} \frac{x^{2n}}{2n-1} \geq 8\gamma \text{Ent}_{\mu}(f). \end{aligned}$$

The ratio tends to 8γ as $x \rightarrow 0$, hence $\rho_0^{\text{opt}} = 8\gamma$.

For the power entropy put

$$D_p(x) = \frac{(1+x)^p + (1-x)^p}{2} - 1 = \sum_{n \geq 1} \binom{p}{2n} x^{2n}.$$

All displayed coefficients are nonnegative for $1 < p \leq 2$, and therefore $x D'_p(x) \geq 2D_p(x)$. Since

$$\mathcal{E}_{\gamma}(f, f^{p-1}) = 2\gamma m^p x [(1+x)^{p-1} - (1-x)^{p-1}],$$

the preceding inequality is exactly

$$4\gamma [\mu(f^p) - \mu(f)^p] \leq \frac{p}{2} \mathcal{E}_{\gamma}(f, f^{p-1}).$$

Again the ratio tends to equality as $x \rightarrow 0$, so $\alpha_p^{\text{opt}} = 4\gamma$. Thus $\alpha_p^{\text{opt}}/\rho_0^{\text{opt}} = 1/2$, proving sharpness.

6 Verification and novelty scope

1. The Stieltjes constants were checked both by differentiation and by numerical quadrature at several non-endpoint values of p .
2. The proof uses only the source’s Assumption 1; no Markov generator, closedness, chain rule, or integral representation of $\mathcal{E}$ is added.
3. The finite-mixture argument explicitly avoids assuming continuity of the abstract bilinear form.
4. The endpoints, the source’s domain condition on f^{p-1} , and exact normalization factors were checked separately.

The local run indexes were searched for the arXiv identifier, exact question, “best constants”, and the mLSI–Beckner comparison. A bounded external search on 11 August 2026 used the exact question and Stieltjes-mixture terms. It found the source and related work on power entropies and curvature, but no explicit solution of this sharp universal-constant question. This is not an exhaustive MathSciNet or citation-graph review, so originality remains a candidate claim pending expert checking.

References

- [1] R. Adamczak, B. Polaczyk, and M. Strzelecki, Modified log-Sobolev inequalities, Beckner inequalities and moment estimates, *J. Funct. Anal.* 282 (2022), 109349; arXiv:2007.10209, doi:10.1016/j.jfa.2021.109349.